

# Townsend Brown

*The notebooks, the instruments, and the question of time — a provenance-and-scorecard paper on Thomas Townsend Brown (1905–1985): what he built, what he measured, what he wrote, and where every "time travel" sentence attached to his name actually comes from.*

draft 0.2 · 2026-08-18 · Osika Innovation AB · Alexander Osika, Claude

**Claim boundary.** This paper is *about* a person and his record; it makes no new physics claim and changes no other paper's verdicts. Its job is to correct a shallowness in our own electrogravitics lane, which had reduced Brown to "the asymmetric-capacitor man whose force is ion wind." That verdict stands ([Electrogravitics](#) §1), but it covers perhaps a fifth of what Brown actually spent sixty years doing. The other four fifths — a thirty-year "sidereal radiation" instrument program, a rock-electricity program, a structure-of-space program, and a family of gravitational-mass hypotheses — are laid out here from the primary record (his own dated notebooks, the 1952 Office of Naval Research audit, his patents and 1975 paper) and scored with the project's three registers: [DOCUMENTED](#) [OPEN THREAD](#) [TRACED TO FABRICATION](#), with [REFUTED](#) for claims that met an independent test and failed it. Method as in [Walter Russell](#) and [Viktor Schauburger](#) §6, under the standing rule that "it is called a myth" is the start of an investigation, not the end — and that the filter cuts both ways.

## 0. The short version

1. **Brown was not a one-idea man.** The released notebooks (Vols. 1, 2, 4; Vol. 3 was never released) contain 243 numbered, dated, sometimes witnessed entries from Oct 1955 to Feb 1977 covering at least seven distinct research programs. The asymmetric capacitor is one of them and, by his own 1973 re-appraisal, the least interesting to him.
2. **Nothing Brown wrote claims time travel.** Not the notebooks, not the patents, not the 1929 article, not the 1952 ONR file, not Project Winterhaven, not the 1975 petrovoltatics paper. What he *did* write about, obsessively, is *time as a*

*coordinate of an unknown signal*: instrument readings he believed varied with solar, lunar, and above all *sidereal* time, plus secular drifts and sudden "glitches." That is a claim about a radiation, not about temporal displacement.

3. **The closest thing to a "time" anomaly in his own hand** is §213 (Avalon, 23 Jul 1975): a Catalina-granite sensor whose chart, he says, *led* Dow-Jones trend reversals by about four days (two days in 1937/39). Five reversals, no statistics, no follow-up published. It is a correlation-hunting note by a man who also proposed a "Magnum Fund" to trade on it, and the ONR auditor who saw the 1937 claim in 1952 found "no convincing evidence." [OPEN THREAD](#) at best — a retrievable-data question, not a physics result.
4. **Every "Brown built / believed in a time machine" sentence traces to two chains, neither of which is Brown**: (a) Moore & Berlitz's 1979 *Philadelphia Experiment* chapter → Bielek's 1990 Montauk time-travel lore; (b) Paul Schatzkin's anonymous e-mail source "Morgan" (2004–05) → *Defying Gravity* serial → *The Man Who Mastered Gravity* (2023) → the Jesse Michels documentary (2024) → Linda Brown's "My Father Was A Time Traveler" clip (2025–26). Schatzkin himself has since written that later insights "challenged the provenance" of Morgan and that he ran with a version of events "however uncertain I was of the veracity"; one of Morgan's checkable claims (an April 1945 Halifax drop into Germany) has been checked against the RAF 38 Group operations record and fails.  
[TRACED TO FABRICATION](#) as stated; see §5.
5. **The two "vacuum proof" documents, read in the original, do not say what the legend says**. The Bahnson notebooks (1957–59) put the famous "100% counterbary" in *air* at 210 kV / 4.6 mA ( $\sim 10$  W per gram, ordinary lifter efficiency), record no pressure on any bell-jar gram figure but one, and carry Brown's own velometer surveys of the airflow; the Montgolfier reports (1957, 1959) put Brown's disc-wire geometry at *null* on a torsion balance at  $5 \times 10^{-5}$  mm Hg and give one uncontrolled  $\approx 1$  g·cm positive on a barium-titanate article (§7).
6. **What is genuinely open** is narrow and specific: the *sidereal harmonic* of his electrometer records was never tested by anyone (Cady tested solar and lunar and found none; he wrote that no test was applied for the sidereal claim); the petrovoltic rocks were never independently instrumented with electrode controls; his 1937 hourly table exists (Cady held it in his hands) and is not public. Those are retrieval and bench tasks with binary outcomes, and §9 states them.

7. **One correction against ourselves:** Brown's notebooks never claim a static vacuum thrust, but his *letters* do (1955 to Hull; 1973 to Rho Sigma: the Paris/Bahnson/GE tests "proved quite conclusively" it operates in high vacuum — notes "not available to me now"). So "Brown never claimed it" is false; "Brown's own records don't support it, and the French record he cited is null on his geometry" is the accurate sentence (§7).
8. **Where he was right** is also worth saying: the in-air force points cathode→anode and scales with leakage current (his own 1973 conclusion, which is the ion-wind mechanism); his 1975 guess that Weber's gravitational-wave "events" were not bar oscillations matched the eventual consensus; his 1957 "dielectric memory" note anticipates electret / ferroelectric non-volatile storage in spirit; his self-potential geothermal prospecting is real geophysics. A person can be right about the shape and wrong about the ingredient — the same verdict our field guide already gives his capacitor.

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## PART I — THE RECORD

### 1. Why we were shallow, and what the primary sources are

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Our [field guide](#) and [bench-floor survey](#) cite Brown for exactly one thing: the 1928 patent, the "gravitator," and the lineage of vacuum nulls that bracket its force. That is the part of Brown that reached the refereed literature (Talley 1988/91; Tajmar 2004; Tajmar & Schreiber 2020; Martins & Pinheiro 2011). It is correct and we do not weaken it here.

But it is a small window. Brown's own record is mostly self-published or family-curated, and it is large. The primary sources this paper stands on:

| Source                                | What it is                                                                                                                                                                                                | Provenance / status                                                                                                                                                                                                                               |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scientific Notebooks<br>Vols. 1, 2, 4 | 243 numbered, dated entries, Oct 1955 – Feb 1977; 310 pp. in the <a href="#">archive.org / rexresearch</a> transcription (Townsend Brown Estate, 2006). Vol. 3 (secs. 123–178, ~1974) "was not released." | Photocopies circulated by William Moore (the <i>Philadelphia Experiment</i> co-author) in the 1980s; family site confirms. Brown's preface (1 Oct 1955): a record book because napkin notes "have ultimately been lost."<br><div>DOCUMENTED</div> |

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|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ONR Pasadena<br>File 24-185<br>(W. M. Cady,<br>15 Sep 1952)                                                                        | 17-page audit of Brown's devices<br>and data, opened 3 May 1952 at<br>Navy request; declassified 1 Oct<br>1952.                                                                                                                                                                                                                                               | The <i>only</i> independent technical audit Brown<br>ever received. Contains the only calibrated<br>numbers on a Brown saucer and the only<br>outside statistics on his electrometer.<br><b>DOCUMENTED</b> |
| Patents                                                                                                                            | GB 300,311 (filed 15 Aug 1927); US<br>1,974,483 (1934); US 2,949,550<br>(filed 3 Jul 1957, 1960); US<br>3,018,394 and 3,022,430 (1962);<br>US 3,187,206 (filed 9 May 1958,<br>1965); US 3,196,296 (1965).                                                                                                                                                     | <b>DOCUMENTED</b>                                                                                                                                                                                          |
| "Anomalous<br>Diurnal and<br>Secular<br>Variations in<br>the Self-<br>Potential of<br>Certain Rocks"<br>(Honolulu, 22<br>Mar 1975) | Brown's one late paper; "publication<br>was withheld." Family site.                                                                                                                                                                                                                                                                                           | <b>DOCUMENTED</b> as a document; unrefereed.                                                                                                                                                               |
| Bahnson–<br>Brown<br>notebooks<br>(Winston-<br>Salem, 1957–<br>59; 55 pp.)<br>and the<br>Bahnson<br>home-movie<br>film             | archive.org (2024); ~1 h silent 8/16<br>mm cut to ~5 min.                                                                                                                                                                                                                                                                                                     | <b>DOCUMENTED</b> ; the 55 pages read in full for<br>\$7.1 (grams, kV, mA in air; no pressure on<br>any vacuum entry but one).                                                                             |
| Gray Barker<br>Collection,<br>Brown files<br>(Clarksburg-<br>Harrison<br>Public Library;<br>20 scans, 166<br>pp,<br>archive.org)   | Brown's 22-page letter to Edward<br>Hull (Nov 1955); the 1946 and<br>c.1948 Lake States Securities<br>brochures marketing his "sidereal<br>radiation" recorder as a stock-<br>market indicator; his 1959–65<br>letters on the Whitehall-Rand<br>collapse; his 1973 letters (via Rho<br>Sigma) and 1975–76 papers with Z.<br>V. Harvalik's lab notes; a c.1936 | Read in full for this paper (§§2, 4.1, 5).<br><b>DOCUMENTED</b>                                                                                                                                            |

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|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                         | Zanesville feature; Moore's c.1979 interview galley.                                                                                                            |                                                                                                                                  |
| Project Montgolfier reports (Sud Aviation, 1 Jul 1957; final, Courbevoie, 15 Apr 1959; Cornillon's copy on archive.org) | Paris 1955–59 tests, in air and vacuum.                                                                                                                         | Read in the original for this paper ( MONTGOLFIER_REPORT_RETRIEVAL_2026-08-18.md ); §7 carries the numbers.<br><b>DOCUMENTED</b> |
| Secondary chains                                                                                                        | Schatzkin (ttbrown.com; 2005–08 serial; 2023 book), Linda Brown, Nick Cook (2001), Moore & Berlitz (1979), Michels (2024), family site thomastownsendbrown.com. | Registered per claim in §5.                                                                                                      |

## 2. The documented life

Dates below are the ones with a document behind them (patent, Navy record, DANFS, archive listing, dated notebook entry, ONR file). Family-timeline items with no independent document are marked.

| Date                | Event                                                                                                                                                                                                                                                     |             |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 18 Mar 1905         | Born, Zanesville, Ohio.                                                                                                                                                                                                                                   | <b>DOC</b>  |
| 1923–25             | One year at Caltech (Millikan unimpressed); Denison University 1924–25. The "Biefeld" of Biefeld–Brown is Denison's Paul Biefeld; Denison stated in 2004 it has no record of the experiments or the association. Biefeld's 1930 affidavit is family-held. | <b>OPEN</b> |
| Aug 1927 / Nov 1928 | GB 300,311 filed / published — "A method of and an apparatus or machine for producing force or motion." Text asserts an unbalanced force from a charge difference.                                                                                        | <b>DOC</b>  |
| Aug 1929            | "How I Control Gravitation," <i>Science and Invention</i> .                                                                                                                                                                                               | <b>DOC</b>  |

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|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Sep 1930 – May 1933 | US Navy: enlisted; NRL Anacostia from Mar 1931 (project on massive high-K dielectrics; the "sidereal radiation electrometer" dates from here by his account); 1932 Navy–Princeton gravity expedition on submarine S-48; 1933 Johnson–Smithsonian expedition, yacht <i>Caroline</i> .                                                                       | <a href="#">DOC</a>                                                                      |
| 1937 / 1939         | Electrometer recording stations, Ohio and Philadelphia. Only the 1937 hourly table survived to 1952.                                                                                                                                                                                                                                                       | <a href="#">DOC</a> (via Cady)                                                           |
| 1938–42             | Recalled as Lt.; Bureau of Ships acoustic/magnetic mine-sweeping design (Oct 1940 – Mar 1941); Atlantic Fleet Radar School, Norfolk (1942). Resigned Sep/Oct 1942 "to escape trial by General Court Martial" (his own timeline's words). Vega/Lockheed radar consultant 1942–45.                                                                           | <a href="#">DOC</a>                                                                      |
| 1946–48             | Lake States Securities Corp. (Cleveland / Laguna Beach) markets Brown's "sidereal radiation" recorder — "Model M-29," 29th since 1923, in a vault in the LA financial district — as a stock-market leading indicator under exclusive licence from the Foundation; brochures claim a ~one-month lead and name Maris (NRL), Biefeld and Miller as endorsers. | <a href="#">DOC</a> (Gray Barker files)                                                  |
| Dec 1947 – 1951     | Hawaii; Pearl Harbor Navy Yard consultant; Adm. Radford's interest; "shift of capacitance mid-point" experiments (Notebook §51).                                                                                                                                                                                                                           | <a href="#">DOC</a> (residence, notebook) / <a href="#">OPEN</a> (demonstration details) |
| 1952–53             | Townsend Brown Foundation, Los Angeles; Project Winterhaven proposal (UMD archives); April 1952 press demonstrations; ONR audit (Cady, 15 Sep 1952) concludes the force is electric wind.                                                                                                                                                                  | <a href="#">DOC</a>                                                                      |
| Oct 1955            | Notebook Vol. 1 begins, Leesburg VA.                                                                                                                                                                                                                                                                                                                       | <a href="#">DOC</a>                                                                      |
| 1955–57             | Paris, SNCASO, Project Montgolfier (Cornillon); funding lost at the Sud-Aviation merger; report 1959.                                                                                                                                                                                                                                                      | <a href="#">DOC</a>                                                                      |
| Oct 1956 – Jan 1957 | Co-founds NICAP (president pro-tem); ousted Jan 1957.                                                                                                                                                                                                                                                                                                      | <a href="#">DOC</a>                                                                      |
| Nov 1957 – 1960     | Bahnson Laboratories / Whitehall-Rand, Winston-Salem NC (Notebook §§57–81 are dated Winston-Salem/Walkertown). "100% counterbary" claimed 11 Jan 1958, in air, tethered. Then a nine-year                                                                                                                                                                  | <a href="#">DOC</a>                                                                      |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                     |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
|             | notebook silence: "October 1958 to October 1967 (9 years) no notes were made."                                                                                                                                                                                                                                                                                                                                                                                    |                                                     |
| 1959–65     | Brown assigns all patents to Whitehall-Rand for stock; the company is "taken over by a group in Meadville, Pa." and collapses; Rand International (Bahamas); Electrokinetics Inc. (Philadelphia, Martin Decker) forecloses on a \$20,000 loan and takes models, equipment and patents; Brown files a federal fraud action (Apr 1965). His words: "greed, bad blood and under-capitalization"; "I really didn't know how ruthless ... profit-minded men could be." | <b>DOC</b> (his 1963/65 letters, Gray Barker files) |
| 1962–68     | Bahnson killed in a plane crash 3 Jun 1964; Homestead FL 1964; Guidance Technology (Santa Monica) 1967.                                                                                                                                                                                                                                                                                                                                                           | <b>DOC</b>                                          |
| 1970–73     | Atherton/Palo Alto (capacitor gravitational-wave bridge, §87, May 1970); Catalina Island 1973 (triboexcitation, the resistance bridges, K- $\mu$ waves, §§89–122).                                                                                                                                                                                                                                                                                                | <b>DOC</b>                                          |
| 1974–78     | Petrovoltaics: Haleakala (1974), Hawaii Inst. of Geophysics vault (1975), Honolulu paper (Mar 1975), Sunnyvale, UC Berkeley Lawson Adit (1976), SRI screen rooms, NASA-Sunnyvale steel chamber (1978); preliminary patent applications (rock battery, "electric clock").                                                                                                                                                                                          | <b>DOC</b>                                          |
| Feb 1977    | Last released notebook entry (§243): "no conclusions can yet be reached."                                                                                                                                                                                                                                                                                                                                                                                         | <b>DOC</b>                                          |
| 27 Oct 1985 | Dies, Catalina Island; buried Avalon.                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>DOC</b>                                          |

**Absent from the notebooks, worth stating.** Across 243 entries Brown never names Biefeld, Bahnson, Whitehall-Rand, Cornillon, SNCASO or Paris. The Winston-Salem entries are dated and placed but the sponsor is invisible. If a session later reads "the notebooks describe the Bahnson vacuum tests," it is wrong; they describe field-shaping theory written *at* Bahnson's lab.

### 3. The seven programs in the notebooks

Read whole, the released volumes are not one project but seven, overlapping in time. Section numbers are Brown's.



### 3.1 Gravitational isotopes and the g/i ratio (§§1–2, 9, 17–18, 23, 44, 90, 95)

The founding hypothesis of Vol. 1 (7 Oct 1955): every element is a mixture of "lighter and heavier isotopes (values of specific gravity)"; gravitational and inertial mass are reciprocal, " $m_g \cdot m_i = E$ ," so a "light" fraction has  $g/i < 1$ . He credits the observed scatter in tabulated specific gravities, Charles F. Brush's 1920s "retardation of g" in complex silicates and Brush's "spontaneous evolution of heat," and proposes "beneficiation" by settling and centrifuging (patent applications were filed; a DuPont heavy-liquid project is described in §16). By 1973 (§95) the Foundation had a "gravitational periodic table" — ytterbium "59% light," silicon 31%, aluminium 30%. In §90 he concedes Dicke's  $10^{-11}$  equivalence bound and retreats to "strong fields would show non-equivalence."

**Status.** REFUTED by physics bounds: Eötvös/Dicke/MICROSCOPE constrain composition-dependent g at  $10^{-13}$ – $10^{-15}$ , twelve to fourteen orders below Brown's clay and Brush's aluminate. Specific-gravity scatter is porosity, allotropy, impurity, working. His 1973 Mason-jar paint-shaker tests (§§90–91: damp Sorrento clay loses  $\frac{1}{4}$  oz in 30 oz on the Avalon post-office scale, "0.8–1%") are inside moisture loss, electrostatic charging and thermal buoyancy; he himself writes the scales "could be in error" and never records the Fisher-balance follow-up he ordered.

### 3.2 Photo-, tribo- and spark-excitation (§§6–14, 20–24, 40, 45–49, 89–96)

Irradiated silicates "may lose weight" and stay warm; freshly ground rock is lightest and recovers on a half-life curve, so "the approximate date of grinding may be determined" (§47); ancient builders may have rubbed sun-baked Nile mud on megaliths (§24). One marginal note reads "Ultrasonic resonance plus laser" (§45). REFUTED by the same bounds; the proposed quartz-capsule-under-UV weighing (§40) was never run.

### 3.3 The vacuum impulse effect (§§42–43, Apr 1956)

The one first-hand vacuum observation in Brown's own hand, and it is worth quoting his framing rather than the lore's: at  $\leq 10^{-6}$  mm Hg "any simple vacuum capacitor will appear to flash as the voltage increases, and, concurrent with the vacuum spark, an impulse force is noted in the direction of the negative to positive." Flash thresholds 30–40, 50–60, 80–90, 130–140 kV; forces "measurable in thousands of dynes." He then names the mechanism himself: "positive conduction, at least as the initiator," "occluded



atoms ... metallic ions ... perhaps even microscopic pieces of metal," reddish anode luminescence, "white star-like spots of considerable brilliance" on the cathode.

**Status.** DOCUMENTED and *explained*: those are cathode spots. A vacuum arc ejects metal plasma at  $\sim 10^4$  m/s and recoils the cathode toward the anode — the vacuum-arc thruster, a mature technology ( $\sim 10\text{--}20\ \mu\text{N}$  per amp of arc). Martins & Pinheiro (Phys. Procedia 20, 2011) map the published Brown-type vacuum impulses onto exactly this; Talley (AFAL 1988, PL-TR-91-3009) found no DC thrust at  $10^{-6}$  torr and forces only during breakdown. Brown's notebook agrees with the nulls: the impulse rides on the spark, not on the static field. That is why our field guide can carry both "Brown saw a force in vacuum" and "there is no static vacuum force" without contradiction — and why the honest verdict on the whole lineage is *refuted as a static effect, open as a transient one*: every positive in the record, Brown's included, is a flash, pulse or breakdown event, which is at once the regime a discharge explains and the only regime our framework's own force law allows ([Electrogravitics](#) §1.1, with the four discriminators — impulse per coulomb, electrode mass, chopper-vs-spark, K-scaling — that would tell them apart). Brown's own §42 already contains one of those controls: the impulse did not appear when the voltage change came from an external chopper, only from the spark.

### 3.4 Field shaping, the toroid, dielectricity (§§56–81, Winston-Salem 1958)

The most engineering-dense stretch: shaped high-flux regions as "a kind of ether pump"; a cathode toroid at the focus of a parabolic anode; the push-pull control grid ("Reversal of polarity does not reverse the direction of the force"); a new fluid "dielectricity — conducted by vacuum, insulated by metals," generated by "dielectromotance," flowing in vortices, producing "dielectric wind"; Jeans's Maxwell-stress bookkeeping (§77) and electrostatic "pressure confinement" for a saucer (§§79–81); the January 1958 analysis of the Adamski "Venusian scout ship" photograph as an engineering exhibit — opened with "This may be a bit of fantasy or it could be significant" and never revisited after 1958.

**Status.** No anomaly. "Dielectric flux" is  $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$ ; motion of a neutral dielectric toward high field is dielectrophoresis (Pohl 1951),  $\propto \nabla E^2$ , polarity-blind, AC or DC — precisely the polarity-independence Brown records; oil "swirl" is EHD plus ion injection, which is Cady's 1952 mechanism supported by Brown's own glass-plate control. He concedes the electric-wind confound in §71 ("difficult to purify the

results"). US 3,187,206 (filed 1958) is this program as a patent — shaped electrodes and graded-K dielectrics — and its "not zero in vacuum" clause is the one Talley contradicted.

### 3.5 The sidereal radiation electrometer → the resistance bridges (§§5, 51, 97–122)

The longest thread of Brown's life and the one our lane had never looked at. Treated in full in §4.

### 3.6 The structure of space: K- $\mu$ waves (§§107–109, 113, 234, 238)

Catalina, Sep 1973: "space, so-called empty space may actually have structure";  $c = 1/\sqrt{K\mu}$  because K and  $\mu$  are "twin energy-storage means" that delay light; regions of differing K,  $\mu$  may propagate as "K waves" and " $\mu$  waves," which "may be instantaneous (infinite velocity) and be truly 'action at a distance' as envisioned by Newton"; a differential-K capacitance bridge and differential- $\mu$  inductance bridge would detect them. Then, in the same entry: "no, on second thought, I take this back. We will run into trouble." Mass is added as a third permeability, m, "gravitational permeability."

**Status.** [OPEN THREAD](#) as speculation, never operated as an instrument beyond the bench bridges of §4.4. Note for this project: it is a polarizable-vacuum picture (variable K,  $\mu$ , hence variable c) — the same family as Puthoff's PV formulation that our [warp paper](#) cites; Brown reached the shape in 1973 with no mathematics behind it. Nothing here concerns time beyond the superluminal-signalling remark.

### 3.7 Petrovoltaics, the gravitocell, qualight (§§179–243, 1975–77)

Treated in §6.

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## PART II — THE INSTRUMENTS

## 4. The sidereal radiation electrometer

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### 4.1 What it was

Brown's own account (§§97, 106, 200; 1975 paper): first model as a Caltech undergraduate, 1923; the NRL 1931 version discussed with Hulburt, Maris, Gunn and Dawson as "a mass and dielectric constant effect"; by 1975 "twenty-nine models." The best hardware description is not Brown's but Cady's, who inspected the Los Angeles instrument in 1952 in a thermostated sub-basement:

- An oil-filled steel tank  $\sim 2$  ft in diameter; a 12-inch, 1-inch-thick disc of six *pine* sectors alternating with six *marble* sectors (low-K vs high-K, low vs high density) on a Bakelite hub, separated by copper strips alternately grounded and connected to a mercury cup; hung on a music-wire torsion fibre; grounded shield  $\frac{1}{2}$  inch away.
- Drive:  $-11.1$  kV smoothed half-wave DC, leakage current 0.45 mA, applied for 30 s every 3 minutes; the disc rotates "several degrees" and the deflection is recorded by a spark on rice paper on a once-a-day drum. Cady: "The apparatus appears to be well engineered and extremely reliable."
- Rest reading 50; deflected  $\approx 80$  in 1952 — but  $\approx 20$  eighteen months earlier, i.e. the *sign of the torque had reversed once, permanently*. Superposed fluctuations of 10–50%.

Brown's own numbers for the 1937 Ohio and 1939 Philadelphia stations (§§98–99, 119): a peak at 16 h sidereal time, resistance maximum about 2 h after the Moon's upper transit, minimum at solar noon, a  $\sim 4$ -day lag after full Moon which he attributes to "capacitance lag" of an Earth of  $\sim 1$  farad. He compares his 1939 Philadelphia curve directly with Dayton Miller's 1926 Mt Wilson interferometer curve, notes both have features at 16 h ST, and records that Miller, shown the electrometer records in Cleveland, called them "inverted" — which Brown in 1973 (§98) accepts: the instrument scale was inverted, so his 16 h ST *peak* "would actually be a thrust minimum."

His fullest description in his own words is the 22-page November 1955 letter to Edward Hull (Gray Barker files): begun at NRL 1932, rebuilt at Penn in 1938, moved to California c. 1943 — two years in a shielded room at Laguna Beach, then a Los Angeles vault  $\sim 30$  ft down under  $\sim 18$  ft concrete-equivalent, two electrostatic shields plus  $\frac{1}{2}$ -inch steel, under oil, held to  $<0.1$  °C, exciter stable to 0.1%, one reading every three minutes,  $\sim 10,000$  a month, about \$20,000 spent. Peaks as he then stated them: solar maximum  $\sim 4$  AM with a secondary  $\sim 4$  PM, minima noon and midnight; lunar  $\leq 5\%$

about 2 h after upper transit; sidereal maxima  $\sim 16$  h and  $\sim 4$  h RA. In the same letter: "There is no net gain of energy from the gravitational field. The speed of spinning varies in a diurnal fashion." And in his 1975 paper: the California recordings of 1944–49 did *not* confirm the eastern lunar effects — "this caused confusion." His 1936 Zanesville press feature already describes a government-built instrument in his parents' basement measuring "the electrical bombardment of the earth by the sun."

#### 4.2 The one independent analysis: Cady, 1952

Cady's §5 is the only outside statistics ever run on these data. He worked from a table Brown handed him, "Hourly Readings of Instrument, 1937," and a second table of the same year against lunar hour angle. His findings:

- Half-monthly means at 01:00 and 13:00 (Table 2) track each other closely through a strong, real rise during spring 1937 (means  $\sim 26 \rightarrow \sim 30$ ) — hence "no conspicuous diurnal effect."
- Hour-of-day means over twelve selected days (Table 3): "no strong diurnal periodicity, with the dubious exception of a peak at 1100 hours."
- Twenty-four lunar-hour-angle means, Jan–Jun 1937 (Table 4): "Their constance is most striking; they show no dependence upon lunar hour."
- Conclusion §5.3: the data "drift significantly" week to week; there is "no evidence of a solar diurnal period"; the data "most emphatically ... deny" a lunar diurnal period. And §8.6, in full weight: *no test has been applied for the claimed sidereal diurnal period*, nor for the Dow-Jones correlation.
- Mechanism (§§8.4–8.5): an electric-wind analogue in oil — ion injection at the copper strips, differential mobility and differential surface leakage of pine versus marble; the permanent sign reversal "attributable to changed surface conductivity of the sectors." Brown's own control — a grounded disc suspended above the sector disc deflected when the sectors were charged, but not with a glass plate interposed — "would seem to corroborate the idea that a swirling of the oil is essential."

Brown's own 1973 re-analysis (§§105, 113) reaches the same mechanism from the inside: the torque is "a direct function of the total current. There is no other energy source"; "Perfectly dry transformer oil produces no torque"; and, of the 1920s gravitators, "It was even believed that the kinetic energy was somehow derived from the

gravitational field!" — the motion is electrical energy → kinetic via leakage current. What he did *not* concede is that the *variation* was mundane; he re-attributed it to a real secular/cyclic variation of electrical resistivity itself, "one of the most significant discoveries in physics" (§101), and rebuilt the instrument as resistor bridges (§4.4).

**Register.** The electrometer's torque: **DOCUMENTED**, mechanism = ion injection / EHD in leaky oil (Cady 1952; Brown 1973 concurs). Its claimed *solar* and *lunar* periodicities: **REFUTED** on the only tabulated year, by the only outside analyst. Its claimed *sidereal* periodicity and the Dow-Jones correlation: **OPEN THREAD** in the strict sense that no one has ever tested them — and Cady said so in writing. The Tables 2–4 in the ONR file are means only; the raw 1937 hourly table is not public.

### 4.3 The comparators

Brown placed his instrument in a family; the family's fate is the fair context.

| Instrument / claim                                  | Sidereal signature claimed                                                      | Where it stands                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dayton Miller, ether-drift interferometer (1926–33) | Apex RA 4h54m, Dec $-70^\circ$ , 208 km/s; minima near 16 h ST                  | Shankland et al. 1955: thermal; Roberts 2006 reanalysis of the raw sheets: not significant, $<6$ km/s. Brown's §98 defence ("Miller observed something") is a fair sentence about a real dataset that did not survive re-analysis.                                                                                                                                                                                        |
| Joseph Weber, resonant bars (1969–70)               | Coincidences peaked in sidereal time toward the galactic centre, $\sim 1661$ Hz | Every independent bar (Garwin, Tyson, Drever, Braginsky, Kafka) null. Brown's §192 (18 Mar 1975, "I am 70 years old today"): the events "are not indicative of cylinder oscillations" but of something "within, or related to the electrical circuit" — a resistance/voltage burst in the strain gauges. The first half matched the eventual verdict; his proposed cause (secondary radiation from the bar mass) did not. |
| Piccardi (chemical precipitation, 1950s–70s)        | Solar/sidereal modulation                                                       | 1987 statistical review: no common long-term pattern.                                                                                                                                                                                                                                                                                                                                                                     |
| Shnoll (histogram fine structure, 1990s–2010s)      | 1436-min and 1440-min recurrences; collimator pointing                          | Rests on expert visual histogram matching; not independently accepted. Closest modern analogue to Brown's "sidereal-time noise" claims, with the same weakness: post-hoc selection.                                                                                                                                                                                                                                       |

|                                                                 |                                                                           |                                                                                                                                                |
|-----------------------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Hodowanec<br>"gravity-wave<br>detector" (1986)                  | Capacitor + op-<br>amp noise with<br>sidereal structure                   | Amplified 1/f noise; unconfirmed.                                                                                                              |
| Cosmic-ray sidereal<br>anisotropy                               | Real; $\sim 0.05\text{--}0.1\%$<br>(Compton–Getting<br>+ galactic dipole) | The one <i>true</i> ground-based sidereal signal — three<br>orders of magnitude below Brown's 10–50% swings and<br>below his documented drift. |
| DAMA/LIBRA;<br>Lorentz-violation<br>clock/cavity<br>comparisons | Sidereal<br>modulation<br>searched explicitly                             | Nulls at present sensitivity; sidereal frequency shifts<br>bounded at $10^{-19}\text{--}10^{-21}$ .                                            |

The pattern is instructive: *every* claimed macroscopic sidereal modulation in a laboratory instrument has failed re-analysis or been left unconfirmed, and the one real sidereal signal is at the  $10^{-3}$  level in particle counts, not in torques or voltages. That is a strong prior against Brown's 10–50% signal being sidereal. It is not a test of it.

#### 4.4 The resistance bridges and the "space compass" (Catalina 1973, §§99–122)

Brown's late-life reinterpretation: the electrometer had really been measuring a *resistivity differential*, and simple electrical resistance "of copper, and possibly all substances, does vary in a secular and possibly cyclic manner" with lunar/solar/sidereal time and with the Earth's absolute motion. He cites Fernando Sanford's 1892 copper-wire report (peaks 29 days apart, "6 days before full moon" by his re-reading), builds Models A-14/15/16 (carbon vs wire-wound resistors, 6–300 V,  $\mu\text{A}$  nulls) and a "Resistance Cross" of buried Stableohm arms, and proposes a semiconducting rod as "a compass for space" pointing at Miller's apex. He records two-week downward drifts,  $\sim 1$  mV "jumps," "spectral bands" by  $V \times R$ , and "glitches" — momentary conductivity losses.

**Status.** DOCUMENTED; never operated with controls; the behaviours listed (voltage coefficient, humidity drift, 1/f and burst "popcorn" noise of carbon-composition resistors) are textbook component behaviour, and his temperature control is a single point (heat to 90 °F, "no change"). Also DOCUMENTED and worth keeping: the resistance-cross concept — a long-baseline differential null instrument — is unobjectionable and was never built beyond the bench.

## 5. The question of time



This is the section the paper exists for. We take it in three layers: what Brown wrote; where the time-travel sentences come from; and the verdict under the project's forensic filter.

## 5.1 What Brown wrote about time — the exhaustive list

We read all 9,196 lines of the released notebooks for it. Every passage a reader might construe as "time" falls in one of six bins:

| Bin                                                                                                                                        | What is actually there                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Sections                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| (a) Time as coordinate of an unknown signal                                                                                                | Solar, lunar and sidereal periodicities and secular drift of electrometer torque, resistance, rock self-potential; "circadian signatures" of sensors; the 16 h / 17 h ST features; a 4-day post-full-Moon lag read as Earth's RC time constant.                                                                                                                                                                                                                                                                                                           | 5, 51, 74, 97–122, 179–243 (dozens)                 |
| (b) Dielectric memory — "The Possibilities of a New Type of Time-Space Data Preservation. A Method of Recording or 'Memory'" (30 Jan 1957) | Despite the title, this is a <i>storage</i> proposal: all recorders to date need an element "moving at a constant rate"; a dielectric may "remember the manner of recharging" as long as it holds charge, via domain progression paced by the data; test with a $\pm 20$ kV capacitor bridge and a Brush galvanometer. It is a rate-encoded electret / dielectric-absorption memory. Not temporal physics.                                                                                                                                                | 50                                                  |
| (c) Absolute motion                                                                                                                        | "Detection of Absolute Motion by Means of Modulated Inertial Mass" (witnessed, 18 Feb 1956): a mass whose inertial mass is modulated should swing along its absolute velocity vector; later, the space compass. Pre-relativistic framing; no time content.                                                                                                                                                                                                                                                                                                | 30–31, 36, 114–117                                  |
| (d) Superluminal signalling                                                                                                                | Notebooks: "K waves may be instantaneous (infinite velocity)"; "Are K waves limited to the velocity of light? ... If not, why should they be limited?" Then, in-line, "I take this back." Earlier and firmer, in the 1955 Hull letter: "electro-gravitic communication" via longitudinal waves whose "speed of propagation would be significantly faster ... than the speed of light" (Pearl Harbor and LA capacitor-bridge tests; Cady 1952 attributed the LA receiver's response to "a chance rectifying contact"). Faster-than-light signalling is the | 107, 109, 183, 238; Hull letter §§ on communication |

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                     |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
|                                    | one place Brown's own claims brush against causality; he never took the step to time.                                                                                                                                                                                                                                                                                                                                                                                                              |                     |
| (e) Dating by decay                | Freshly ground rock is lightest, so "the approximate date of grinding may be determined"; megaliths' weight recovery dated to "70,000 to 200,000 years ago."                                                                                                                                                                                                                                                                                                                                       | 3, 47               |
| (f) A signal that leads the future | §213, 23 Jul 1975: "a correlation may be found between sidereal radiation and the stock market. Even as early as 1937 such a correlation seemed to exist"; now a Catalina granite on a Triplett meter shows "a phenomenal correlation with the DJ Averages," lag radiation → DJ "about 4 days" (2 days in 1937/39); "Five reversal of trend have, so far, taken place. The probability of this is astronomical"; "No explanation seems to exist"; a "Magnum Fund" prospectus under the Foundation. | 213 (and Cady §5.1) |

Bin (f) is the only place Brown's own hand puts a physical signal *ahead* of an event, and it is the one bin a reader could honestly call a "time" anomaly. So it deserves care:

- It is not new in 1975 — and it was once a business. The Gray Barker files hold two Lake States Securities Corp. brochures (©1946 booklet; c. 1948 tri-fold, "Sidereal Radiation Foreruns the Stock Market") marketing the recorder as a leading indicator with a claimed lead of about *one month*; the 1937 "break" and 1939 "rise" charts are the exhibits. Cady recorded in 1952 that Brown claimed the correlation for the 1937 data ("it has been claimed that a chi-square analysis has been performed, with positive results") and that he, Cady, "has seen no convincing evidence." So the claimed lead is one month (1946–48), two days (1937/39 as recalled in 1975), and four days (1975): three incompatible lags for one signal, which is what a post-hoc alignment looks like. Brown restated the thesis in his own 1976 paper (radiation ↔ optimism/pessimism ↔ stock prices).
- Five trend reversals in eight weeks is not "astronomical" — daily market series reverse trend constantly, and any drifting sensor chart can be aligned to some of them by choosing the lag. Brown supplied no pre-registration, no lag-fixed scoring, and, in the released volumes, no follow-up after July 1975.
- Even taken at face value, a signal that *leads* a human aggregate by four days is not precognition or time travel; it is a claim that both the rock and the market respond

to a common driver with different lags. Brown says as much in his own framing ("incoming radiation covers a broad spectrum").

Register for bin (f): [OPEN THREAD](#), and a shallow one — decidable by retrieval of the 1937 table and the 1975 Catalina chart, both of which existed and one of which Cady physically examined. Everything else in the list is either instrumentation (a), an engineering idea (b), or speculation he withdrew (d).

## 5.2 Where the time-travel sentences come from

Two chains, fully traceable, neither passing through anything Brown wrote:

**Chain 1 — the Philadelphia Experiment.** Jessup / "Allende" letters (1955–56) → Berlitz & Moore, *The Philadelphia Experiment* (1979), whose chapter 10 attaches Brown because he was a Navy officer working on magnetic mine-sweeping in 1940–41 (true) → Al Bielek (1990) adds Montauk and time travel, with Tesla and von Neumann as the "team" and Brown only via Berlitz & Moore. Moore planted a paragraph in a draft; Brown returned it "essentially correct" — and a long-time associate reports that phrase was Brown's private code for a hole "big enough to drive a truck through." Schatzkin himself writes "TPX is a myth"; Cook's source "Marckus" called it "a neat bit of misinformation." One thing must be added for fairness to the record: in Moore's own c. 1979 interview galley (Gray Barker files) Brown is described as declining to discuss it rather than denying it — "ask him about something that happened in 1925 and he talks your head off," but on this: "if Naval Intelligence has it locked up, you're not going to get it" — and Moore's 1984 update mentions testimony sealed until "a year and a day after his death" (Brown died Oct 1985). That is Moore's account of an oblique old man, from the author with the largest stake in the story; it is not a document, and it does not move the register. [TRACED TO FABRICATION](#) — this is the Bennewitz pattern of MIC \$25, a real man's real Navy record used as the seed of a manufactured story.

**Chain 2 — "Morgan."** Paul Schatzkin's serialised biography *Defying Gravity: The Parallel Universe of T. Townsend Brown* (2005–08) and its book form *The Man Who Mastered Gravity* (2023) carry the theme via an anonymous e-mail correspondent, "Morgan," who claimed to have known Brown in the 1960s and who wrote, of Brown's work as described in Cook's *Hunt for Zero Point*: "It's a f—ing time machine." Morgan's Christmas-1965 Homestead anecdote has Brown answering a question about time travel with "Yes, I believe it is possible. In your lifetime." Morgan is also the sole source of the "Caroline

Group" — a Stephenson/BSC-flavoured private network said to have worked on "a time machine ... traveling between dimensions." Chapter 20 of the serial, in which Schatzkin himself speculates that bending space bends time, is labelled speculation by its author.

What has since happened to Chain 2:

- Cook confirmed his own source ("Marckus") was British and met in person; Schatzkin's was e-mail only. Schatzkin: "Maybe not a word I've written is true."
- Morgan's one checkable wartime story — an April 1945 Halifax NA337 drop into Germany — was checked against RAF 38 Group operations records by a forum researcher: the aircraft's sorties were Denmark (aborted) and Norway (ditched in Lake Mjøsa); no Germany sortie was possible. **REFUTED** on record.
- Schatzkin's 2023–24 "Outtakes-37" states that later insights "have challenged the provenance" of Morgan and the "O'Riley/Boston/Twigsnapper" persona, and concedes he ran with a version "however uncertain I was of the veracity." His collaborator adds the "MIB" testimony "may not all be exactly 100% accurate."
- Downstream: the Jesse Michels documentary (Oct 2024; audited in `MICHELS_BROWN_DOCUMENTARY_AUDIT_2026-08-13.md` — 109 minutes, zero measured data) and Linda Brown's 2025–26 "My Father Was A Time Traveler!" clip inherit Chain 2 whole. Linda's own journals show "Morgan" first as a real boyfriend, later merged with the correspondent; she also reports a 1955 three-saucer encounter and a "shared dream-visitation," and describes sceptics as "agents." That is a family witness paying a real cost — the filter's honest-witness criterion — but it is a witness to a *persona*, not to a device or a document.

Provenance map, in one line: *Jessup/Allende 1955 → Moore & Berlitz 1979 → Bielek 1990 // Cook 2001 → Schatzkin/Morgan 2004–08 → book 2023 → Michels 2024 / Linda 2025–26.* Brown's paper trail contributes zero links.

### 5.3 Verdict under the filter

"Brown figured out time travel": **TRACED TO FABRICATION**. The claim's every appearance is downstream of (1) a book its own co-author's associate disowned and Schatzkin calls a myth, or (2) an anonymous e-mail persona whose one testable claim fails the archival record and whose sponsor has publicly withdrawn confidence in it. There is no witness at cost to a

device, no document, no drawing, no notebook line. By the §26/§27 forensic test this is the "anonymous packet" case, exactly as the Haunebu papers were in the Schauburger paper.

**"Brown worked on time":** true only if reworded to what he actually did — thirty years of instruments read against sidereal time, plus one dielectric-memory idea, plus one signal-leads-market note. Reworded, **DOCUMENTED** as a program and **OPEN THREAD** as to whether any of it was real, with the solar/lunar parts **REFUTED** by Cady.

**The filter cutting the other way:** a wrongly-dismissed observation is rescued by evidence, and here the evidence that could rescue Brown's sidereal claim exists in principle — the 1937 hourly table Cady held — and has never been examined for the sidereal harmonic. That is the honest residue, and §9 turns it into a task.

## 6. Petrovoltaics — the rock program (1974–78)

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### 6.1 What Brown claimed

Family-site definition: "the spontaneous rectification and storage of radiation in the form of electrical potential by certain materials," neither piezo- nor pyroelectric. Silver or copper electrodes plated on opposite faces of flat granitic/basaltic rocks give DC potentials "up to 700 mV," typically 60–115 mV in the Hawaii notebooks (Waikiki rock ~10 cm, 60 mV; +Kuhio-Beach sand core 63 → 115 mV; Koolau basalt ~400 mV, the best material), with "systematic diurnal variations" (low 6–7 AM, high 6–8 PM), "sudden random spikes or bursts" (the "glitches": 21 Dec 1974 and 7 Jan 1975 at Haleakala; 4 May 1976 at Berkeley — granite off-scale at 97 mV, at "approx 17 h sidereal time, at approx upper meridian transit of the galactic center. This may or may not be a coincidence") and long-term secular drift. Sensors extended to BaTiO<sub>3</sub>, PZT, glycerin–litharge, tungsten carbide in carnauba, ceramic "cartwheel" and electrolytic capacitors, which "behave like rocks." The 1975 paper cites Weber's galactic-centre claim as its frame and posits "an energy influx of possibly extreme penetrability."

The theory built on it (§§200–208, "the gravitocell"): incident gravitational radiation ("qualight," made from starlight crossing cosmic fields) is absorbed in proportion to mass, knocks charge carriers loose, and self-polarised rocks rectify it; bias of 45 V raises the diurnal signal ~10×; a 10 kV "self-sustaining gravitoelectric generator" is sketched.



Pyramids as gravitic-flux concentrators (§223) and biological effects (§224) follow as explicit speculation.

## 6.2 Controls he says he ran, and what is missing

Brown's list is not nothing: Haleakala at 10,000 ft; sea level; the Hawaii Institute of Geophysics seismic vault (long-term drift only, diurnal suppressed — his own §211); UC Berkeley Lawson Adit, 200 ft of overburden ("200% increase vs surface"); NASA-Sunnyvale 90-ton cobalt-steel chamber (1978); SRI screen rooms; enclosures held to  $\pm 0.1$  °C; "concurrent bursts" 80 km apart; and the statement that humidity, atmospheric electricity, geomagnetism, cosmic/gamma/neutrino flux "have as yet provided no answers." What is missing is what would decide it: the electrode control. Two plated metal films on a moist ionic conductor form a galvanic cell whose potential is exactly hundreds of millivolts; the self-potential geophysics literature says electrode effects must be modelled explicitly; a dawn-minimum / evening-maximum diurnal is the classic thermal-lag/humidity signature; "steel framing reduces readings" is what a high-impedance electrostatic pickup does; and Brown's own last entry (§243) concedes "the emf developed would surely be temperature dependent, and this, it appears, is true." No electrode-swap, no dry-inert-atmosphere, no shorted-Faraday-cage null is recorded.

## 6.3 Replications and citations

One contemporaneous look exists: Z. V. Harvalik's February 1975 comments and lab notes (Gray Barker files) — a 20-channel logger, and the finding that 50–500 M $\Omega$  *carbon resistors* reproduce the rock diurnal (minimum  $\sim 7$  AM, maximum  $\sim 6$  PM) with 5–50 mV glitches, larger at 10,000 ft. That either supports an ambient signal common to rocks and resistors, or — the reading the resistors' known temperature and humidity coefficients favour — a shared environmental driver; Harvalik asked for a Co-60 test that we find no record of. Otherwise none peer-reviewed. Laurie & Salisbury (c. 1998–2008) saw mV-scale *changes* in various rocks with rate-of-change converters and interpret them artistically; an amateur page (now dead) reported a persistent small voltage and "cannot rule out a chemical reaction or some kind of thermal effect." A search for citations of the 1975 Honolulu paper returns only the family site and unrelated self-potential geophysics. The mainstream neighbour of Brown's diurnal claim — tidal influence on self-potential measurements — exists and does not cite him.



**Register.** **DOCUMENTED** claim with real dated notebooks and a real 1975 paper; **REFUTED** as an energy source ( $10^{-1}$  V into  $>10^8 \Omega$  is pW–nW, never near "self-sustaining"); mainstream account (electrode electrochemistry + moisture + thermal lag) is sufficient and was never excluded by a published control; **OPEN THREAD** strictly on whether any *sidereal* component survives such controls — untested by anyone, and cheap to test (\$9).

## 7. The vacuum ledger, restated in Brown's own terms

Because Brown's name is the vacuum claim's flag, it is worth lining up what *he* said against what was measured, so no reader has to reconcile them later.

| Source                                                           | Vacuum statement                                                                                                                                                                                                                                                                                                                                                                                                     | Register                                                                                                                                                                                                    |
|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Brown, Notebook §§42–43 (Apr 1956)                               | Impulse rides on the vacuum spark; anode ions and metal fragments named as initiators.                                                                                                                                                                                                                                                                                                                               | DOCUMENTED; = cathode-spot recoil                                                                                                                                                                           |
| Brown, Notebook §67 (Jan 1958)                                   | Force $\propto$ K and m; "In vacuum, with a K of unity and m = 0, the force should be minimum."                                                                                                                                                                                                                                                                                                                      | DOCUMENTED — his own theory predicts near-null in vacuum                                                                                                                                                    |
| Brown, Notebook §113 (Sep 1973)                                  | The gravitator "moved in the neg to pos direction when charged. This is what happened, even in vacuum." No pressure, no number.                                                                                                                                                                                                                                                                                      | Self-report, un-numbered                                                                                                                                                                                    |
| Brown, letters — to Hull (Nov 1955); to Rho Sigma (Feb/Apr 1973) | 1955: "the tests in Paris last Summer proved this. We were able to go up to 150 kv." 1973: vacuum tests at Paris 1955–56, Bahnson 1957–58, GE Space Center King of Prussia 1959, $10^{-6}$ mm Hg, 70–220 kV, "proved quite conclusively that the apparatus would indeed operate in high vacuum" — but "these notes were never published and are not available to me now." Also 1973: the Navy "rejected the research | Self-report; <b>this is where Brown himself does claim vacuum operation</b> — in correspondence, from memory, without notes; the French balance he cites was, we now know, null on his geometry (row below) |

|                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
|                                                                                                                                                   | proposal" on the electric-wind objection.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                         |
| Cady / ONR (1952)                                                                                                                                 | Efficiency of the air devices ~1.4% (0.3% overall); thrust "presumably ... zero in a vacuum"; ~7 g at 30 kV falling to ~1 g with the corona wire shrouded; calculated electric-wind thrust "about twice the observed" — "satisfactory in order of magnitude."                                                                                                                                                                                                                                                                                                                                                                                                          | MEASURED (the only calibrated Brown-saucer numbers)                                                     |
| Montgolfier / Cornillon (1957 report; 1959 final report) — read in the original 2026-08-18, receipt<br>MONTGOLFIER_REPORT_RETRIEVAL_2026-08-18.md | 1957 bell jar ( $2-5 \times 10^{-5}$ mm Hg): discs "always moved toward the positive pole," motion often flash-driven, stopped by a grounded screen, 2–4 $\mu$ A at 40 kV, unmeasurable; Brown's ~100 suggested variations during his second stay produced no effect above bell-charging noise; Cornillon's own hypothesis = "anodic effect" (particle emission). 1959 torsion balance (1.48 m chamber, $5 \times 10^{-5}$ mm Hg, $\pm 50-55$ kV): Brown disc-wire article <b>null</b> ; two other articles null; plexiglas capacitor 3–4°; BaTiO <sub>3</sub> -loaded capacitor 30° $\approx$ 1 g·cm, reversing with polarity; no current/breakdown/lever-arm record. | OPEN THREAD, now numbered — null on Brown's own geometry; one uncontrolled positive on a high-K article |
| Bahnson Labs notebooks (Winston-Salem, Nov 1957 – Apr 1959; 55 released pages, read for this paper — §7.1)                                        | Bell-jar entries give grams and kV/mA but <b>never a pressure</b> except once ( $7 \times 10^{-5}$ mm Hg, 3 Nov 1957 — which Bahnson himself attributes to coulomb forces: "electric wind effect was eliminated at this vacuum but the coulomb forces are very great"). Brown's 17 Apr 1958 "critical test" entry (concentric arcuate                                                                                                                                                                                                                                                                                                                                  | DOCUMENTED artefact; no controlled, pressure-stated vacuum datum                                        |

|                                                                                            |                                                                                                                                                                                                                           |                     |
|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
|                                                                                            | electrodes with guard rings, in vacuum, "results are positive") carries no kV, mA, grams or pressure. Bahnson later asks whether a jar result "may be partially or wholly accounted for by the rebalancing of the Brush." |                     |
| Talley, AFAL/PL (1988, 1991)                                                               | $10^{-6}$ torr: no thrust under DC; forces only at breakdown / pulsed PZT.                                                                                                                                                | NULL                |
| Tajmar (2004); Canning/NASA (2004); Tajmar & Schreiber (2020, $\pm 0.3$ mg); Tajmar (2024) | No thrust without current; corona wind in air.                                                                                                                                                                            | NULL                |
| Martins & Pinheiro (2011)                                                                  | Vacuum "anomalies" = vacuum-arc thruster physics.                                                                                                                                                                         | DERIVED explanation |

This is the same ledger as [Electrogravitics](#) §3, with two additions: Brown's own theory (§67) and his own observation (§42) both place the static vacuum force at or near zero; and the Montgolfier reports, read for this paper, put Brown's own disc-wire geometry at *null* on the sponsors' 1959 torsion balance while giving one uncontrolled positive on a barium-titanate article. The lore's "Brown proved it works in vacuum" is asserted by Brown in letters (1955, 1973 — from memory, without notes) but by neither his notebooks nor the French report; what the French report does say is that a high-K dielectric "makes the phenomenon more vigorous" — which is the one sentence in the whole 1950s vacuum record that a builder could act on, and it points at the dielectric, not at Brown's capacitor.

### 7.1 The Bahnson notebooks, read (Nov 1957 – Apr 1959)

Fifty-five scanned pages of the three Bahnson-lab notebooks are on archive.org; we read every page for this paper. Most are in Agnew Bahnson Jr.'s hand; about a dozen are Brown's own neat entries; J. Frank King Jr. witnesses. What they establish, in the writers' own words:

- **The "100% counterbary" of 11 Jan 1958 was in air.** The entry: 7:33 PM, the 35-inch stitched-bamboo canopy with the 18-inch foil half-ball, "rig weighs approx 98 grams," lifted its own weight at 210 kV and 4.6 mA — about 970 W,  $\approx 10$  W per

gram, which is ordinary EHD-lifter efficiency — witnessed by Bahnson, King, Brown and one other. Later runs of the same family: 60 g at 225 kV/0.75 mA (2.8 W/g), 180 g at 230 kV/1.95 mA (2.5 W/g). Nothing on the page separates it from ion wind plus attraction to the room; no vacuum equivalent exists in the notebooks.

- **Brown himself mapped the airflow.** His entries of 24 Jul – 8 Aug 1958 are velometer surveys inside the bi-arcuate anode: "air flow is generally outward through all the tubes," about 50 ft/min out of the small end, cathode-shape-dependent pressure "signatures," and — 29 Aug — velocities "computed at 100 KV ... approx. 40 meters/sec." That is the ion-wind mechanism, measured by Brown, in Bahnson's lab.
- **Bahnson's own doubts are on the page.** 10 Feb 1958: after being "amazed" at lift with no ground wire, "the lift must be ... a coulomb attraction of the induced negative charge of the dielectric"; 3 Jun 1958, a hissing cone whose thrust reverses at 90 kV: "It looks like transport effects are working against it," margin note "Ionization seems to fight it"; a conference outline (10 Jun 1958) lists the possible reasons for thrust as electric wind/transport, attraction to environment, interaction with the lab field, and the ground lead; Teller, via Griggs, is recorded as being "in the coulomb camp" and asking for proof it is not transport or environment attraction — a criterion the notebooks never record being met. By late 1958 the agenda reads "electrohydrodynamics ... airfoil ... plasma."
- **The vacuum grams have no pressures.** Three numeric bell-jar entries — 5 g at 95 kV/0.033 mA (10 Jan 1958), "38 gr ... in vacuum(?)" at 140 kV/0.055 mA (4 May), 19 g vs 32 g at 100 kV/ $\sim$ 0.03 mA (Brown, 30 Jun) — record no pressure; the currents (30–80  $\mu$ A at 100–150 kV, i.e. several watts) are what soft vacuum or surface leakage look like, and the jar tests used foil wrapped around the outside of the glass. Brown's own control (1 Jul 1958): with the lead to the upper shield removed, "<1 g at 150 KV ... slight charging of upper shield." Both bell jars fractured in August 1958.

Read as Brown would want them read, the Bahnson pages show a competent program that measured what it could, wrote down its own confounds, and never closed them — not a hidden vacuum proof. The one line the legend leans on (Brown, 17 Apr 1958: "The results are positive and show substantial lifting forces") is the only vacuum entry in the set with *no* number attached.

## 8. Scorecard

| # | Claim                                              | Brown's own numbers                                 | Independent test                                              | Verdict                                                           |
|---|----------------------------------------------------|-----------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------|
| 1 | In-air force on asymmetric capacitor               | 1927–58 patents; ~7 g at 30 kV (Cady's measurement) | Cady 1952; Tajmar 2004; Canning 2004                          | <b>DOC</b> real force; ion wind. Brown 1973 concurs on mechanism. |
| 2 | Static thrust in vacuum                            | none in his hand; §67 predicts minimum              | Talley; Tajmar & Schreiber; Tajmar 2024                       | <b>REFUTED</b> (as coupling bound; see field guide §3)            |
| 3 | Vacuum impulse at spark                            | onset 30–40 kV; "thousands of dynes"                | Talley (breakdown only); Martins & Pinheiro                   | <b>DOC</b> , explained (arc recoil)                               |
| 4 | Gravitational isotopes / g varies by material      | Yb "59% light"; clay –1% on shaking                 | Eötvös-class bounds $10^{-13}$ ; Brush's NBS/GE checks failed | <b>REFUTED</b>                                                    |
| 5 | Dielectromotance / dielectric wind / field shaping | none                                                | —                                                             | No anomaly (D-field, dielectrophoresis, EHD)                      |
| 6 | Electrometer torque                                | 0.45 mA, –11 kV; deflection 50→80; sign reversal    | Cady 1952                                                     | <b>DOC</b> ; ion injection in oil                                 |
| 7 | Electrometer solar / lunar period                  | noon min; +2 h lunar max                            | Cady 1952, 1937 table                                         | <b>REFUTED</b> ("most emphatically")                              |
| 8 | Electrometer sidereal period (16 h ST)             | 1937/39                                             | none — "no test has been applied"                             | <b>OPEN</b> (retrievable)                                         |
| 9 | Signal leads Dow-Jones by 2–4 days                 | 1937/39; 1975 five reversals                        | Cady: "no convincing evidence"                                | <b>OPEN</b> , weak; not precognition even if true                 |

|    |                                                                      |                                     |                                                          |                                                                           |
|----|----------------------------------------------------------------------|-------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------|
| 10 | Resistance of conductors varies with sidereal time / absolute motion | Models A-14/15/16; drifts, mV jumps | none; Lorentz-violation clock bounds $10^{-19}$ as prior | <b>OPEN</b> as stated; component-noise account sufficient                 |
| 11 | K-μ structure of space; superluminal K-waves                         | none                                | —                                                        | Speculation; PV-family shape; withdrawn in-line                           |
| 12 | Petrovoltaic self-potential                                          | 60–700 mV; diurnal; glitches        | none peer-reviewed                                       | <b>DOC</b> ; galvanic/thermal sufficient; <b>OPEN</b> on sidereal residue |
| 13 | Gravitocell / qualight / pyramid focus                               | none                                | —                                                        | Speculation, labelled so by him                                           |
| 14 | Weber events are not bar oscillations                                | §192, 1975                          | Garwin et al. 1972–74                                    | Right conclusion, wrong cause                                             |
| 15 | Dielectric "time-space" memory                                       | §50, 1957                           | —                                                        | Engineering idea; anticipates electret/FeRAM in spirit; not time physics  |
| 16 | Adamski scout-ship analysis                                          | §§58–63, 1958                       | —                                                        | Hedged by him; dropped after 1958                                         |
| 17 | Philadelphia Experiment participation                                | —                                   | Navy record 1940–42; Moore/Berlitz provenance            | <b>FABRICATION</b>                                                        |
| 18 | Time machine / "Caroline Group"                                      | —                                   | Morgan e-mails; RAF ORB refutation; Schatzkin retraction | <b>FABRICATION</b>                                                        |

## 9. What this project takes forward

The residue is small, specific, and cheap. Each item is a bounded task with a binary outcome, in the same spirit as the field guide's §6.



1. **Retrieve the 1937 hourly table.** Cady's ONR file reproduces only means (Tables 2–4, as images). The full "Hourly Readings of Instrument, 1937" and its lunar-hour-angle companion existed in Los Angeles in 1952 and presumably passed to the estate (papers went to San Antonio per the family timeline). If retrieved: fold by *sidereal* hour with the drift removed (Cady's spring rise is a 15% ramp), score the 16 h ST amplitude against a shuffled-day null. This is the single test Cady said had not been applied. Also retrieve the July 1975 Catalina chart against the DJIA with the 4-day lag fixed *before* looking. Outcome: either the sidereal claim gets its first number, or Brown's central lifelong claim is closed on his own data.
2. **Add a "Brown channel" to the sidereal-folding control of the field guide's bench program.** Our five-control discriminator already includes sidereal folding of the drive/force record. Mounting one plated-electrode basalt and one carbon-composition resistor bridge on the same logger, with an electrode-swapped twin, a shorted twin, and a dry-nitrogen twin, costs nothing and answers \$6 and \$4.4 in the same run: if the controls track the live sensors, the diurnal is galvanic/thermal; if a sidereal residue survives all three, that is a real datum. Pre-register the amplitude threshold at the cosmic-ray anisotropy level ( $10^{-3}$ ) as the only physically motivated prior.
3. **Do not import Brown's structure-of-space (§3.6) as support for anything.** It is a shape without a calculation; the PV-family resemblance is resemblance. The [warp paper](#) and [gravity paper](#) own that territory.
4. **Keep the two fabrication verdicts load-bearing.** Any future session that finds "Brown, time travel, Caroline Group" in a source should route it here rather than re-litigating; and, symmetrically, any session that finds a *document* — a drawing, a witnessed page, an archive box — must reopen \$5, because the verdict is on provenance, not on plausibility.
5. **Update the sibling papers** so that "Brown" no longer means only "capacitor": one sentence and a link in [Electrogravitics](#) §1 and [Antigravity Devices](#) §4 pointing here for the notebooks, the electrometer, and the time question.

## 10. A closing word about the man

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The notebooks show a careful, self-critical, unlucky experimenter, not a charlatan. He dated and witnessed entries; wrote "correction," "I take this back," "the scales could be

in error," "possibly 'to absurdity'," "no conclusions can yet be reached"; noticed the ion-wind confound in his own oil (§71) and the current-dependence of his own torque (§105); saw through Weber before Weber's peers did; and kept, for fifty years, records of an instrument that nobody but one Navy physicist ever analysed. He was wrong about gravitational isotopes, wrong about dielectricity as a substance, wrong — on the only tabulated year — about the Sun and Moon in his electrometer, and right that his capacitor pointed the direction it did and scaled with the current it drew. He never wrote a word about time machines. The version of him that did is a character in other people's books, and the man deserves to be scored on his own pages. That is what this paper does; the one test he was owed and never got — the sidereal fold of 1937 — is now on our list.

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Draft 0.2, 2026-08-18 (0.1 same day; 0.2 adds the Bahnson notebooks, the Montgolfier reports read in the original, and the Gray Barker files). Registers follow the project filter (Documented / Open thread / Traced-to-fabrication; Refuted where an independent test exists). No number in this paper is new; every number is Brown's, Cady's, or a cited author's. Not deck material.